UFCFK1-15-0 - Fundamentals of data science 25feb_1

Take home assessment 40% (PSA)

Documentation 2

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Section : A

GIT : https://github.com/999-ztj/FODS_assignments

Assignment 2

Question 1

Program Description:

This program uses a function to calculate sum, difference, product, and quotient.

Screenshot:

```
Enter first number: 1
Enter second number: 2
Sum: 3
Difference: -1
Product: 2
Quotient: 0.5
```

Question 2

Program Description:

This program performs mathematical operations using separate functions.

Screenshot:

```
Enter first number: 1
Enter second number: 2
Addition: 3
Multiplication: 2
Division: 0.5
Floor Division: 0
Exponentiation: 1

Process finished with exit code 0
```

Question 3

Program Description:

This program checks whether a number is an Armstrong number.

Screenshot:

```
Enter a number: 153
Armstrong number
```

Question 4

Program Description:

This program counts the frequency of numbers in a list.

Screenshot:

```
{1: 1, 2: 2, 3: 1}
{4: 3, 5: 1}
{7: 2, 8: 1, 9: 1}
```

Question 5

Program Description:

This program sorts student names in reverse alphabetical order.

Screenshot:

```
Enter names separated by comma: Jenith,Sara
Sorted names: Sara Jenith
```

Question 6

Program Description:

This program searches for a country in a list.

Screenshot:

```
Enter countries separated by comma: Nepal,Christmas_islands,Georgia,Mongolia
Enter country to search: Christmas_islands
1
```

Question 7

Program Description:

This program counts characters in a sentence excluding spaces.

Screenshot:

```
Enter a sentence: God is good God is great
{'G': 2, 'o': 4, 'd': 3, 'i': 2, 's': 2, 'g': 2, 'r': 1, 'e': 1, 'a': 1, 't': 1}
```

Question 8

Program Description:

This program removes duplicates and sorts numbers.

Screenshot:

```
Enter numbers separated by space: 2 2 2 2 2 2 2 2 2 5 6 7 1 2 1 3 4 566 66 4 4 3
Result: [1, 2, 3, 4, 5, 6, 7, 66, 566]

Process finished with exit code 0
```

Question 9

Program Description:

This program stores details of 3 movies in a dictionary.

Screenshot:

```
Enter movie title: Marry my Husband
Enter director: Saranshika Kc
Enter release year: 2026
Enter rating: 9
Enter movie title: Friren
Enter director: Jenith Gurung
Enter release year: 2026
Enter rating: 10
Enter movie title: Maze runner
Enter director: Sara kc
Enter release year: 2024
Enter rating: 3

Title: Marry my Husband
Director: Saranshika Kc
Year: 2026
Rating: 9

Title: Friren
Director: Jenith Gurung
Year: 2026
Rating: 10

Title: Maze runner
Director: Sara kc
Year: 2024
Rating: 3

Process finished with exit code 0
```

Question 10

Program Description:

```
This program is a dice guessing game.
```

Screenshot:

```
Guess the dice value (1-6): 5
😐 Close!Dice was: 4
```

```
Guess the dice value (1-6): 1
❌ Wrong! Dice was: 5
```

```
Guess the dice value (1-6): 6
😊 Correct!
```